REPO-ALIGNED REVISION

Symphonix-Health CRAFTS agentic intelligence architecture

Revised against the local Symphonix repositories. The prior draft treated several repos as inaccessible; this edition replaces that constraint with code and specification evidence.

Working interpretation: Symphonix-Health is best described as governed, networked clinical intelligence infrastructure. It is not evidenced as a generic AGI platform. "AGI" can remain a research horizon term, but the architecture that exists is a bounded clinical agent network with identity, consent, routing, safety, and evidence gates.

CRAFTS status: CRAFTS is used here as an analytical lens: Culture, Relations, Agency, Funding or resource governance, Technology, and Structure. It is not a runtime subsystem in the codebase.

CORRECTION

Repo access is no longer treated as a blocker. GHARRA, Nexus A2A, BulletTrain, Bridge SDK, Prompt Engine, CSAA, SignalBox, CAID, and Patient360 provide direct evidence.

REMAINING GAP

The missing object is not a persona registry. It is a single cross-plane projection binding persona, skills, tools, IAM, safety, budgets, consent, and evaluations.

- BulletTrain data plane
- GHARRA trust registry
- Nexus A2A
- Bridge SDK contracts
- CSAA safety cases
- SignalBox attestation

WHAT CHANGED

Primary revisions from the incomplete source report

> The source PDF's strongest limitation was its claim that local repositories were unavailable. This revision treats the public narrative as secondary and uses the local repos as the control surface.

Source draft position	Repo-aligned correction	Evidence status
Architecture is "spec-verified and interface-verified, not fully code-verified."	Core claims are now code/spec verified across BulletTrain, GHARRA, Nexus, Bridge SDK, Prompt Engine, CSAA, SignalBox, CAID, and Patient360.	**closed** Local repositories are readable.
Unified persona registry may be missing.	GHARRA has a `PersonaRegistry`; Bridge SDK now also has a `SuperpersonaContract` exchange shape for human/AI agent identity, skills, tools, IAM scopes, safety, eval evidence, route salience, and runtime budgets.	**reframed** Human/AI agent capability framework adoption across every sibling system is still partial.
Specialty persona completeness is not verified.	Local persona code includes psychiatrist, dietitian, pharmacist, and nurse practitioner examples with toolsets and FHIR role codes.	**verified** Completeness of every specialty remains a testable inventory task.
Compute power should be added as a new CRAFTS dimension.	The code supports a narrower term: resource governance. Bridge SDK defines per-persona and per-patient budgets; Bevan and Prompt Engine route model work through governed gateways.	**refined** No evidence of an infrastructure ownership layer.
Computational intuition can be treated as a realized layer.	The repos show planning, reflection, route salience, provider fallback, capability discovery, and safety gating, but no single global "intuition" service.	**contracted** Bridge SDK has a contract-level `IntuitionDecision`; global clinical deployment remains partial.

Wording change

Use "networked clinical intelligence" for the current architecture. Reserve "networked superhuman clinical intelligence" for a carefully bounded thesis claim, not for code readiness.

Diagram change

The revised diagrams should put BulletTrain, Bridge SDK, GHARRA, Nexus, Prompt Engine, Bevan, CSAA, SignalBox, CAID, sibling/domain systems, and the human/AI agent capability framework into explicit layers instead of showing a generic agentic stack.

ARCHITECTURE SPINE

Repo-backed architecture layers

The codebase supports a layered clinical intelligence architecture. Data, identity, reasoning, execution, and evidence are separate concerns.

External substrate	Health systems, national rails, payers, pharmacies, labs, imaging, ambulance workflows, and external model providers are dependencies, not proof of autonomy.	`Kenya SRS`
		`Bridge adapters`
		`BulletTrain`

Sibling domain systems	Pharmacy, clinical pathways, ambulance/EMS, insurance eclaims, ERP, supply-chain ERP, HMIS, provider portal, citizen portal, appointments, genomics, and Patient360 are the domain execution surfaces.	`pharmacy-system`
		`clinical-pathways`
		`HMIS + portals`

Data plane	Cross-system data movement belongs in BulletTrain. Bridge SDK translates external protocols and projects consent, provenance, FHIR tasks, and audit records.	`BulletTrain` `symphonix-bridge-sdk`
Model and reasoning plane	Prompt Engine and Bevan route model calls through governed gateways with policy prompts, schema validation, PHI boundaries, fallback, and output checks.	`prompt-engine` `BulletTrain/bevan_llm`
Trust and route plane	GHARRA provides zero-trust discovery and trust metadata. Nexus validates agent records and runs A2A route admission and task lifecycle.	`global-agent-registry` `nexus-a2a-protocol`

Human and AI agent framework	Human roles and AI agent personas/superpersonas carry skill packs, tool packs, clinical frameworks, authority scope, safety constraints, evidence duties, and outcome targets.	`GHARRA personas`
		`Bridge superpersona`
		`Nexus + sibling tools`

Governance and evidence plane	CSAA, SignalBox, CAID, RBAC, audit stores, HITL coordination, and real-service test rules decide whether an agent path can move beyond shadow or assisted operation.	`csaa`
		`signalbox-mcp`
		`caid-agent`

CRAFTS MAPPING

CRAFTS after code verification

CRAFTS remains useful, but each dimension should map to real components and contracts. The categories below avoid treating social theory terms as deployed services.

Culture

Clinical norms, safety language, national constraints, consent posture, and model behavior rules.

`Prompt Engine policy prompts; CSAA; Kenya SRS; design invariants.`

Relations

How agents, humans, tenants, patients, and systems can see, trust, and route to one another.

`GHARRA; Nexus route admission; Bridge consent/provenance; Patient360 context radius.`

Agency

Bounded capacity to propose, draft, classify, route, summarize, or escalate clinical and operational work.

`68 persona definitions; 25 Nexus agents; Bridge typed intents; HITL workflows.`

Funding and resource governance

Resource ceilings, model routing, concurrency, queues, and cost-sensitive routing, not a broad funding theory.

`Bridge concurrency budgets; Bevan routing/fallback; workflow metrics.`

Technology

The actual data, protocol, model, agent, and observability systems that make the network executable.

`BulletTrain; Bridge SDK; Prompt Engine; Bevan; SignalBox; CAID.`

Structure

Governance boundaries, role permissions, safety cases, audit duties, and release gates.

`RBAC; GHARRA ABAC; CSAA hard stop; real-service evidence rules.`

Mechanism claim: Symphonix changes the clinical decision environment by making data movement, identity, persona scope, model routing, safety, and evidence visible and governable. The mechanism is the governed network, not a free-floating agent.

Dimension	Mechanism class	Boundary condition
Culture	Norms and policy prompts shape acceptable clinical behavior.	Policy text is not enough without tests and safety cases.
Relations	Trust metadata and consent projection decide which exchanges can happen.	GHARRA remains no-PHI; PHI routing must stay local.
Agency	Persona-scoped intents and tools make agent action bounded.	Legal or clinical effect needs human approval unless evidence says otherwise.

PERSONA AUDIT

The human and AI agent framework gap moved

> The old report treated a unified persona registry as likely missing. Local code changes that conclusion: GHARRA has a consolidated persona registry. The live gap is cross-plane projection from human and AI agent personas into skills, tools, frameworks, IAM, authority, safety, budgets, consent, evaluations, and outcome obligations.

Verified persona surface

- GHARRA `PersonaDefinition` includes key, label, category, summary, toolset, SOPs, domains, FHIR role code, and avatar variants.
- `PersonaRegistry` exposes the consolidated persona catalog.
- The code comments state that the module consolidates three persona systems, 68 persona definitions, country-aware avatar selection, and capability scoping.

Specialty examples now verified

- Psychiatrist persona exists with clinical toolset and FHIR role code.
- Dietitian persona exists with nutrition toolset and FHIR role code.
- Pharmacy personas and nurse practitioner personas are present.

Plane	Current evidence	Cross-plane risk
GHARRA	Persona definitions, avatar variants, capability scoping, FHIR role codes, no-PHI trust boundary.	Registry does not by itself prove runtime authorization.
Nexus	25 clinical agents, 68 personas, six IAM groups, A2A route admission, JSON-RPC task lifecycle.	Needs generated contract alignment with GHARRA personas.
Bridge SDK	Typed agent intents, safety class, FHIR Task projection, persona delegation token fields, superpersona contracts, agent-card output.	Needs generated framework, skill, and tool-pack adoption across siblings.
Patient360 and BulletTrain	Persona-centered patient context, FCG personas, HITL orchestration, clinical workflow governance.	Needs evidence that UI roles, data radius, and backend contracts agree.

CONTRACT TARGET

Human and AI agent superpersona contracts now exist

Bridge SDK now implements a `SuperpersonaContract` exchange shape. The contract should be treated as the capability framework for human and AI agents: persona identity, skills, tools, clinical frameworks, authority, safety, evidence, and outcome binding. Adoption across GHARRA, Nexus, SignalBox, Prompt Engine, Patient360, and sibling systems is still partial.

persona_id
`GHARRA PersonaRegistry`

agent_kind
`human | AI | team`

human_role
`FHIR role code; IAM group`

skill_pack
`persona toolset; SOPs`

framework_pack
`clinical pathway; SOP; policy`

tool_pack
`Bridge SDK capabilities`

authority_scope
`IAM; consent; HITL policy`

safety_class
`Bridge AgentContract; CSAA`

jurisdiction_pack
`Kenya SRS; GHARRA variants`

consent_projection
`Bridge consent_projection.py`

budget_limits
`Bridge concurrency.py`

route_admission
`Nexus A2A; GHARRA metadata`

model_policy
`Prompt Engine; Bevan router`

eval_pack
`SignalBox; CAID; scenario YAML`

outcome_pack
`service + strategic metrics`

audit_pack
`Bridge audit; BulletTrain audit store`

Contract field	Exists locally?	Needed revision
Human/AI agent identity and clinical role	**yes** GHARRA persona definitions and Nexus IAM groups.	Generate one canonical ID and alias table consumed by every sibling repo.
Skill, tool, and framework authorization	**partial** Bridge SDK `SuperpersonaContract` now projects skill and tool policy; explicit framework-pack consumption is not universal.	Compile and publish allowed skill packs, tool calls, and clinical/operational frameworks per agent, task type, tenant, and jurisdiction.
Safety and evaluation binding	**partial** Safety classes, CSAA, SignalBox, CAID, and Bridge eval-pack references exist.	Make deployment fail when a persona lacks a passing safety case, SignalBox/CAID evidence, and sibling route tests.
Budget and compute limits	**yes** Bridge concurrency budgets and superpersona runtime budget policies exist.	Report saturation telemetry and require budget fields in each sibling contract projection.
Outcome binding	**partial** Outcome telemetry and evidence systems exist, but are not yet one mandatory agent-capability gate.	Bind each human/AI agent framework to immediate outputs, service outcomes, strategic outcomes, and innovation/invention evidence.

RESOURCE GOVERNANCE

"Compute power" is better modeled as resource governance

The source report proposed compute power as a possible extra CRAFTS dimension. The repos support a narrower and more useful architecture term: resource governance.

What is code-backed

- Bridge SDK has per-persona and per-patient execution budgets with saturation behavior and limit dimensions.
- Bevan and Prompt Engine route model calls through governed gateways rather than letting every sibling call arbitrary models directly.
- GHARRA documentation preserves a no-PHI boundary; PHI model routing belongs inside local Bevan-controlled paths.
- HITL queue metrics and workflow gates provide operational back-pressure.

What is not evidenced

- No local evidence was found for an estate-level GPU ownership or data-center control layer.
- No single service currently declares all persona budgets, model budgets, queue limits, and safety gates as one contract.
- Therefore the diagram should not imply compute sovereignty beyond the code-backed routing and budget controls.

1 **Task arrives**
Persona, tenant, patient, task, consent, and jurisdiction context are identified.

2 **Budget check**
Per-persona and per-patient limits decide whether the task can execute now.

3 **Route choice**
Prompt Engine and Bevan choose local or external model paths under policy.

4 **Safety gate**
CSAA, HITL policy, output validation, and consent rules constrain action.

5 **Evidence**
Audit, SignalBox scenarios, CAID checks, and metrics decide readiness.

Revised CRAFTS wording: keep Funding as a social and procurement context, but add "resource governance" as the deployed system dimension that CRAFTS must inspect for agentic healthcare.

REASONING LAYER

Computational intuition is contract-backed, not a global service

The idea is useful when defined as auditable pre-planning: salience detection, route choice, likely framework selection, risk spotting, and tool-pack selection before deeper execution. Bridge SDK now has an `IntuitionDecision` inside superpersona contracts; the repos still do not show one global clinical intuition service.

Candidate function	Repo-backed evidence	Required to claim it as a layer
Fast salience detection	Prompt Engine detects clinical/PHI patterns; Bridge SDK records route salience in `IntuitionDecision`.	Cross-sibling salience tests across pathways, pharmacy, claims, EMS, and portals.
Framework selection	Persona definitions, SOPs, toolsets, clinical safety classes, and route-salience records provide the vocabulary.	A deterministic framework-pack selector that records why a pathway, SOP, policy, or reasoning frame was chosen in live clinical workflows.
Route and model choice	Bevan and Prompt Engine support governed LLM routing, fallback, and output checks.	A route decision trace bound to persona, consent, and safety case.
Risk spotting	CSAA can hard-stop unresolved residual risks; Bridge SDK has safety classes and clinical safety cases.	Risk classifier evidence tied to SignalBox/CAID scenarios.

Recommended diagram label

"Contracted salience and routing heuristics" is more accurate than an unlabeled intuition layer for the current architecture.

Recommended implementation

Extend the existing contract record with input context, selected persona, selected tools, model path, risk class, evidence pack, and outcome feedback.

Do not imply autonomous clinical diagnosis from this layer. The verified behavior is bounded recommendation, drafting, routing, classification, summarization, and escalation under human and safety governance.

NESTED DIAGRAM

Corrected nested architecture

The nested stack below replaces a generic agentic diagram with repo-backed layers. Each layer depends on the layers below it and is constrained by safety and evidence above it.

Learning and readiness	SignalBox scenarios, CAID reflection and structured memory, Prompt Engine optimization, CSAA risk-case updates, real-service tests, outcome telemetry, and false-positive/false-negative checks.
Action and workflow	Patient360 plus sibling systems: pharmacy, clinical pathways, ambulance/EMS, insurance eclaims, ERP, supply-chain ERP, HMIS, provider portal, citizen portal, appointments, genomics, lab, imaging, and assistant journeys mediated through BulletTrain and Bridge SDK.
Guardrails and trust	CSAA safety cases, GHARRA trust metadata, Nexus route admission, RBAC/ABAC, HITL coordination, consent projection, audit, and break-glass controls.
Reasoning and routing	Prompt Engine, Bevan LLM router, policy prompts, provider fallback, schema validation, local PHI routing, and output validation.
Human and AI agent framework	GHARRA human roles and AI personas, Nexus agents/IAM, Bridge superpersona contracts, skill packs, tool packs, framework selection, authority scope, safety class, runtime budgets, FHIR role codes, and outcome obligations.
Context and data	BulletTrain integration doctrine, FHIR R4, OpenHIE, X12/EDI, terminology, longitudinal patient context, consent, provenance, and audit stores.
External substrate	National systems, payer rails, provider systems, pharmacies, labs, imaging networks, identity providers, and external model APIs under explicit adapter contracts.

> The diagram should be read bottom-up for dependency and top-down for governance. No agent bypasses the data, consent, trust, safety, sibling-system, and evidence layers. Layer 6 should be labelled "Symphonix-Health mechanism core."

> **Diagram additions:** add the human/AI agent capability framework, add bounded rationality to the essence-critical problem domains, and add innovation and invention to strategic outcomes. These are not cosmetic labels; they need outcome metrics and evidence before they are treated as achieved.

ROADMAP

Roadmap after repo access was restored

The roadmap no longer starts with "restore repository access." It starts with turning already-present surfaces into generated contracts and executable gates.

Workstream	Status	Next action
Persona inventory	partly closed	Generate a normalized persona inventory from GHARRA, Nexus, Patient360, Bridge SDK, and BulletTrain role files. Flag aliases and missing mappings.
Human/AI agent framework	implemented core	Bridge SDK implements the contract shape; now generate/adopt agent-kind, skill-pack, tool-pack, framework-pack, authority, safety, evidence, and outcome fields across GHARRA, Nexus, SignalBox, Patient360, Prompt Engine, and sibling systems.
Safety as CI gate	partial	Make CSAA unresolved risk, missing SignalBox scenario, missing CAID check, or missing real-service journey fail readiness.
Resource governance	partial	Project Bridge concurrency budgets and Bevan routing limits into persona contracts and telemetry dashboards.
Architecture catalog	gap	Create a cross-repo generated map of core services, sibling systems, routes, persona roles, data flows, PHI boundaries, and evidence artifacts.
Outcome learning loop	partial	CAID, SignalBox, Prompt Engine, CSAA, and Bridge SDK have loop pieces; bind clinical outcomes to controlled workflow, prompt, safety, and contract updates.
Sibling test gates	failing	Repair reduced-matrix requirement-ID drift and requirement-count drift in CAID-generated siblings before claiming all sibling capabilities are tested.

1

Inventory

Extract persona, route, contract, safety, and scenario surfaces from each repo.

2

Normalize

Resolve aliases, role codes, IAM groups, tenant scope, and jurisdiction variants.

3

Generate

Emit human/AI agent capability contracts and architecture catalog artifacts.

4

Gate

Require safety, real-service, and scenario evidence before readiness claims.

5

Learn

Feed verified outcomes, bounded-rationality misses, and innovation/invention evidence into controlled contract and policy updates.

VERIFICATION DISCIPLINE

Rules for future diagrams and reports

Future architecture documents should use these checks before claiming readiness, coverage, or agentic maturity.

Evidence rules

- Use local repo code/specs as the control source before public pages or narrative reports.
- Distinguish code evidence, requirement evidence, scenario evidence, and inference.
- Do not count reports, dashboards, or generated matrices as verdicts without source checks.
- Run false-positive and false-negative checks for disputed coverage claims.

Clinical readiness rules

- Use real seeded service paths for internal systems; do not replace reachable internal services with mocks as readiness evidence.
- No agent path should bypass consent, provenance, audit, PHI boundaries, HITL policy, route admission, and safety cases.
- For backend contracts, UI evidence alone is incomplete unless network/backend evidence is also captured.
- Break-glass, emergency, and legal-effect workflows need explicit human control and audit evidence.

Claim type	Minimum evidence	Bad shortcut to avoid
Persona exists	Registry entry, role code, alias mapping, IAM mapping, and runtime route or UI use.	Counting a name in one markdown file.
Agent can act	Typed intent, allowed tools, consent, safety class, route admission, budget, audit, and tests.	Inferring action from a model prompt alone.
Workflow is ready	Real seeded data path, backend and UI checks, SignalBox/CAID evidence, and safety case status.	Scenario-only pass without service evidence.
Architecture is integrated	Cross-repo contract map, route map, service ownership map, and data/provenance flow.	One diagram with no generated ledger.

SOURCE LEDGER

Repo evidence used for this revision

Repo/file	Lines	Evidence used
symphonix-health-docs/strategy/agent-first.md	9; 19-21; 33-42; 54-59; 193-209	Agent-first sibling roles, API boundaries, CSAA/GHARRA/Nexus control model, discoverable/reachable/grounded/gated/observable/reversible requirements, HITL patterns.
symphonix-health-docs/strategy/bullettrain-integration-doctrine.md	11-23; 61-100; 117-119	BulletTrain as the required cross-system data movement plane; AI agents may recommend/draft/classify/route but cannot mutate sibling systems outside authorized auditable mediation.
BulletTrain/README.md	33; 43-69; 121-130; 183-219; 318-380	160+ microservices, FHIR/OpenHIE/X12/terminology support, SignalBox/FCG/governance, RBAC, HITL, Bevan, Patient360, context assembler, workflow orchestrator, adapters.
global-agent-registry/README.md	26; 93-119; 139; 165-166; 259-268	GHARRA zero-trust discovery/routing/trust registry, no-PHI boundary, ABAC, consent, FHIR R4, break-glass, agent cards, and dashboard.
global-agent-registry/src/gharra/core/personas.py	4-18; 187-199; 288-296; 524-533; 545-583; 625-633; 970-984	Consolidated persona registry, 68 persona definitions, country-aware avatar selection, capability scoping, persona fields, psychiatrist, nurse practitioner, pharmacy, dietitian, and `PersonaRegistry`.
nexus-a2a-protocol/README.md	26-30; 43; 137; 187-200; 222; 263	Trusted identity, AI Gateway, JSON-RPC, JWT, task lifecycle, 25 clinical agents, 68 personas, six IAM groups, identity endpoints, persona/IAM docs.
nexus-a2a-protocol/docs/gharra_integration.md	5; 14; 18-21; 228-231	Nexus consumes GHARRA trust/routing metadata, validates records, and does not replace GHARRA discovery or BulletTrain orchestration.
symphonix-bridge-sdk agent and superpersona contracts	agent_contract.py 1-13; 275-310; superpersona_contract.py 1-8; 79-102; 167-190; 224-292; tests/test_superpersona_contract.py 17-103	Agent Execution Contract, typed intents, FHIR Task conversion, GHARRA agent-card output, human/AI superpersona contract fields, skill and tool policy, route salience, runtime budgets, GHARRA/A2A/MCP/audit exports, and direct tests.
symphonix-bridge-sdk/src/bridge_sdk/concurrency.py	1-4; 23; 57-86; 102-182	Per-persona and per-patient execution budgets, saturation, dimensions, and limits.
symphonix-bridge-sdk/src/bridge_sdk/consent_projection.py	16-23; 95; 123-148	Consent projection travels on persona token and task, including jurisdiction and stale-consent checks.
prompt-engine/docs/REQUIREMENTS.md	344-363; 643-670; 950-956; 1026	Agentic mode, persistence/tool-use/planning, safety policy prompt elements, reflection/error recovery, PHI and clinical pattern detection.
prompt-engine/src/prompt_engine/optimizer/workflow.py	4-11; 227	Every LLM call routes through the BulletTrain API gateway hub and schema validation.
csaa; signalbox-mcp; caid-agent; patient360-assistant	CSAA 90-94, 149; SignalBox v2 217-229; CAID 245-249; Patient360 README 55-61, 128-132	Safety scope and hard-stop behavior, scenario evidence, FP/FN audit discipline, Patient360 clinical assistant and persona/context expectations.
Sibling-system audit	nested-crafts-diagram-cross-check.md; sibling README/test evidence	Pharmacy, clinical pathways, ambulance/EMS, eclaims, ERP, supply-chain ERP, HMIS, provider portal, citizen portal, appointment, genomics, and Patient360 must be visible in the diagram, along with the human/AI agent capability framework that binds personas to skills, tools, frameworks, and outcomes; several matrix gates currently fail.